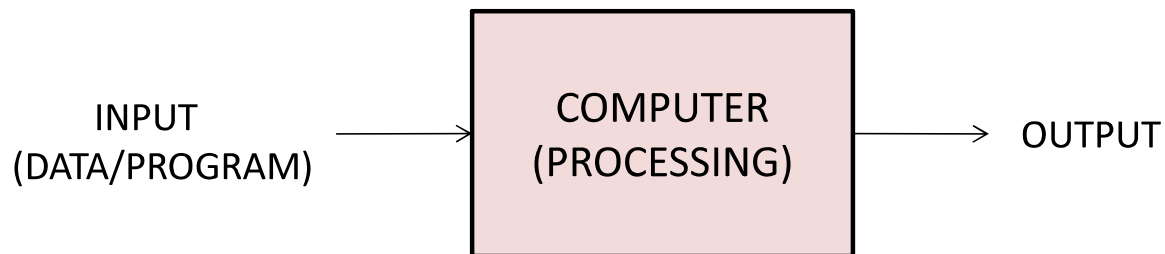


Introduction to Components of a Computer System

What's a Computer?

A **computer** is an *electronic machine* that

- takes input from the *user/memory*,
- processes the given input according to a **program** (set of instructions)
- and generates output in the form of useful information.



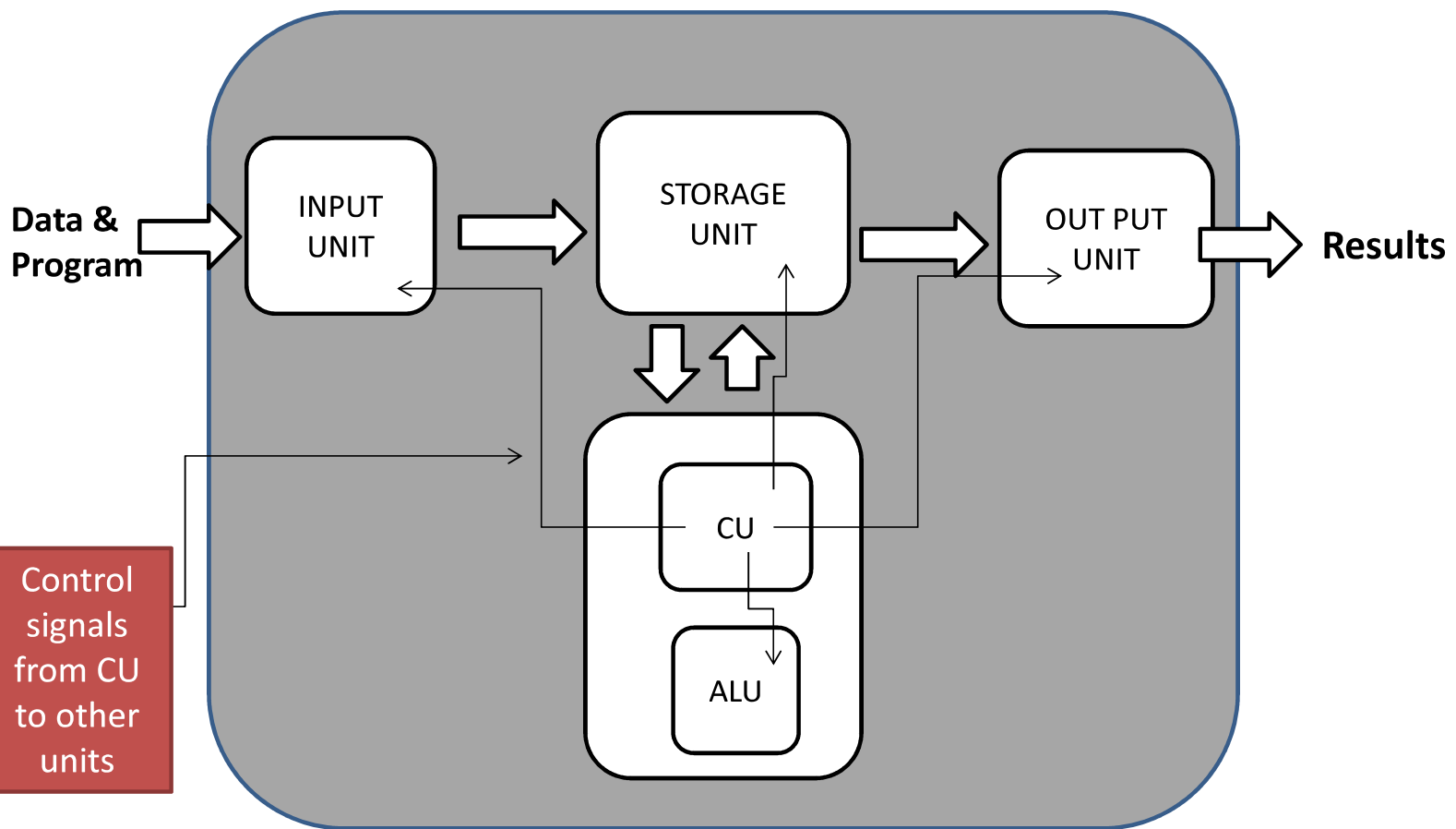
Basic Operations of a Computer System

- Taking Input
 - Processing/Controlling
 - Generating Output
 - Storing data and Programs
-

Functional Components

1. Input Unit
2. Output Unit
3. Storage Unit
4. Processing Unit
 - Control Unit
 - Arithmetic and Logic Unit

Computer System: Block Diagram



1. INPUT UNIT

Performs the following functions :

- 1. It accepts or reads the data and program (set of instructions).*
- 2. It converts these instructions and data in computer acceptable form.*
- 3. It supplies the converted instructions and data to the computer system for further processing.*

– Keyboard, Mouse, Scanner, Digital Camera etc..

2. OUTPUT UNIT

Performs the following functions :

- 1. It accepts the results produced by the computer which are in coded (binary) form.*
- 2. It converts these coded results to human acceptable form.*
- 3. It supplies, the converted results to the outside world.*

-monitor, printer, plotter etc..

3. STORAGE UNIT

Performs following functions:

- 1. It stores the data and the program (set of instructions).*
- 2. It holds the intermediate results during processing.*
- 3. It stores the final results of processing before they are passed onto the output unit.*

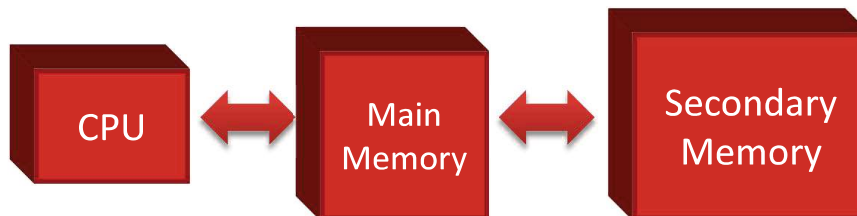
4. Central Processing Unit (CPU)

- **The major sections of a CPU are:**
 - (i) Arithmetic and Logic Unit (ALU),**
 - (ii) Control Unit (CU).**
- **ALU:** Performs arithmetic and logic operations such as addition, subtraction, multiplication and division: AND, OR, NOT, EXCLUSIVE-OR Operations.
- **Control Unit:** The control unit is the most important part of the C.P.U. as it controls and co-ordinates the activities of all other units such as ALU, memory unit, input and output unit.
 - *Gets the instructions from the memory unit.*
 - *Decodes the instructions.*
 - *Generates control signals accordingly for other units.*

Storage Unit

Memory can be categorized into two types:

1. Main Memory or Primary Memory
2. Secondary Memory or Auxiliary memory



Difference b/w Primary and Secondary memory

PRIMARY MEMORY

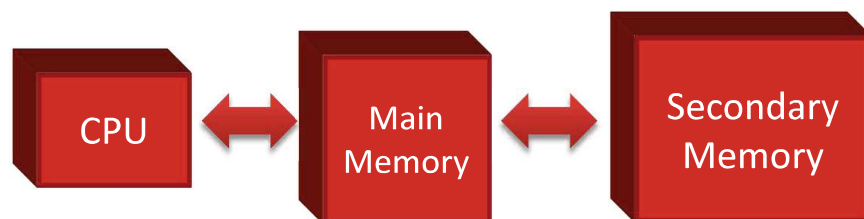
1. Primary memory can be directly accessed by CPU.
2. The data & programs are loaded into the main memory prior to the execution.
3. It has limited storage capacity.
4. Its access speed is fast.
5. It is expensive.

Example: RAM, ROM.

SECONDARY MEMORY

1. Secondary memory can not be directly accessed by CPU.
2. It is used to store data and program for future references.
3. It has large storage capacity as compared to primary memory.
4. Its access speed is slower as compared to primary memory.
5. It is cheaper as compared to primary memory.

Example: HDD, CD, DVD....



Difference b/w RAM & ROM

READ/WRITE MEMORY (RAM)

1. It is read and write memory i.e. Data can be read as well as can be written into this memory.
2. It is volatile in nature. When the power is switched off, all data stored in it is washed out.
3. It is used by CPU to store data, programs and intermediate results

READ ONLY MEMORY (ROM)

1. It is read only memory ie. Data can only be read from this memory. Data can not be written into this memory.
2. It is non-volatile in nature. If the power is switched off, it retains its data.
3. It is used to store BIOS (Basic input/output system). BIOS is a special software that controls the startup process of a computer.

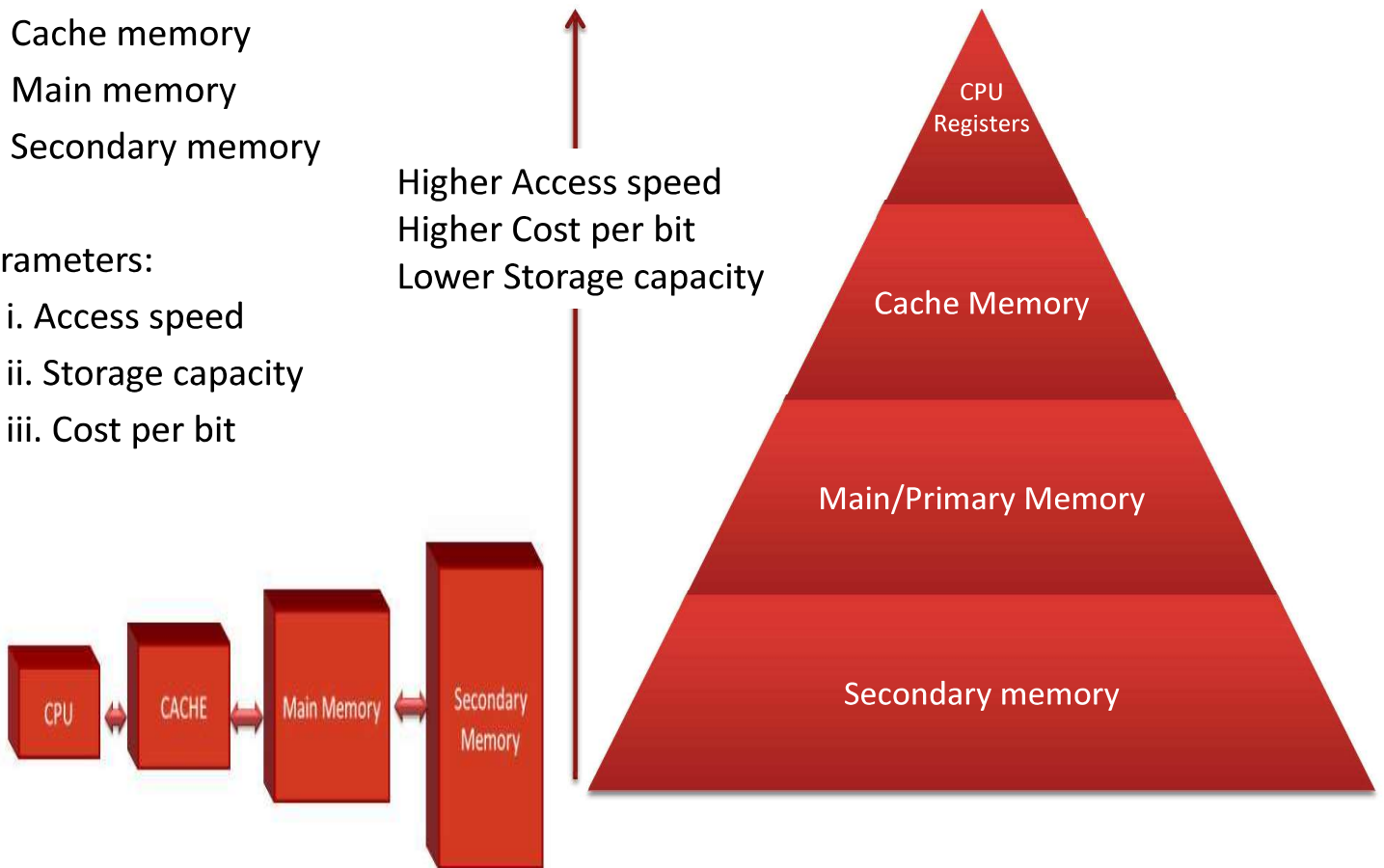
Memory Hierarchy

- CPU Registers
- Cache memory
- Main memory
- Secondary memory

Parameters:

- Access speed
- Storage capacity
- Cost per bit

Higher Access speed
Higher Cost per bit
Lower Storage capacity

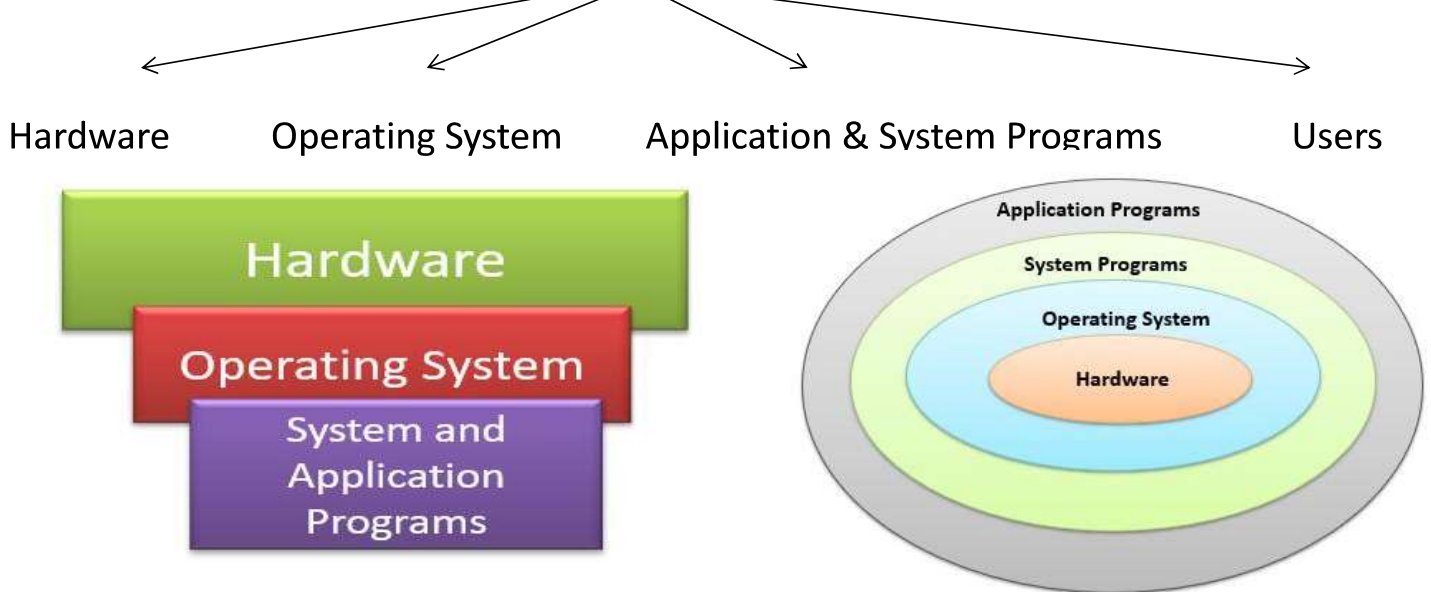


Operating System

Operating System

- Operating System is a software, which makes a computer to actually work.
- A system program that manages hardware.
- The OS organizes and controls the hardware.
- OS acts as an interface between the application programs and the machine hardware.

Computer System



Operating System

Functions of Operating System:

1. Memory Management:

- allocation of memory to the running programs
- removes the programs from memory when it completes

2. Process management:

- processing the jobs, deciding on the job scheduling technique

3. Device management (Resource Allocation):

- allocating the input and output devices to the running processes and de-allocating them when the processes are over.

4. File Management:

- managing the file system in terms of where the files are stored, their status and memory locations.
- opening and closing the file

Types of operating systems:

1. Batch System
2. Multiprogramming
3. Time-sharing or Multi-tasking
4. Multi-processor

Algorithm and Flowchart

Algorithms

Definition:

In computer programming terms,

“An algorithm is a set of well-defined instructions to solve a particular problem.”

It takes a set of input(s) and produces the desired output.

“A logical sequence of steps for solving a problem.”

“A plan of solution for a problem”

Algorithms: examples

2. To multiply two numbers (*MORE PRECISE***)**

- 1: START
- 2: Input A and B.
- 3: Product = $A * B$.
- 4: Print Product.
- 5: END.

Step 1: START

Step 2: Input/Scan the Values of A and B.

Step 3: PRODUCT = $A * B$.

Step 4: Print the value of PRODUCT.

Step 5: END.

Algorithms: examples

3. To calculate average of three numbers

1. START
2. Input/Scan A,B and C.
3. $AVG = (A+B+C)/3$
4. Print AVG
5. END

Algorithms: examples

Examples:

4. To find larger between two distinct numbers

1. START
2. Input A & B.
3. IF $A > B$
 Print A
 ELSE
 Print B
4. END

Examples:

4. To find larger between two distinct numbers

1. START
2. Input A & B.
3. IF $A > B$ goto step 4
 otherwise goto step 5
4. Print A
5. Print B
6. END

Algorithms

Characteristics:

1. Finiteness
2. Definiteness
3. Inputs
4. Output
5. Effectiveness

Algorithms

Characteristics:

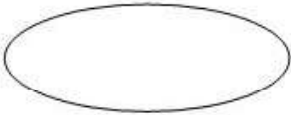
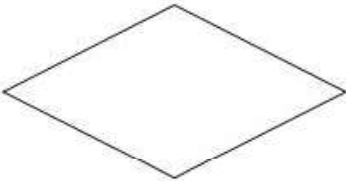
- 1.Finiteness:** An algorithm must terminate after a finite number of steps and further each step must be executable in finite amount of time.
- 2. Definiteness (no ambiguity):** Each steps of an algorithm must be precisely defined.
- 3.Inputs:** An algorithm has zero or more but only finite, number of inputs.
- 4.Output:** An algorithm has one or more outputs
- 5.Effectiveness:** An algorithm should be effective. This means that each of the operation to be performed must be sufficiently basic that it can, in principle, be done exactly and in a finite length of time, by person using pencil and paper.

Flow-Chart

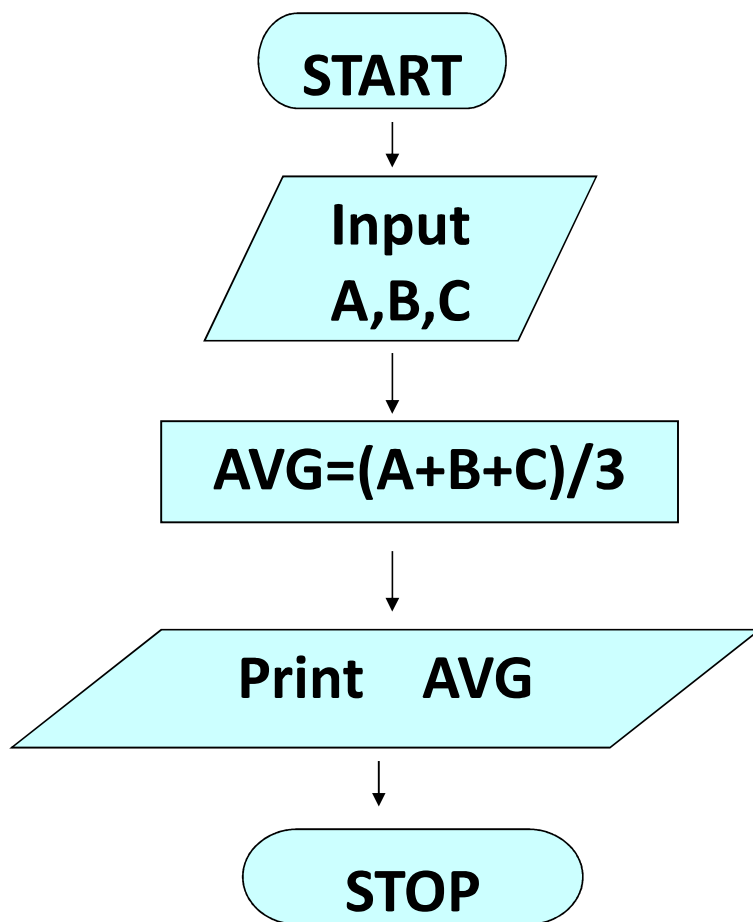
The Flowchart

- A pictorial (graphical) representation of the sequence of operations.
- Different symbols are used to draw each type of operation.

Flowchart Symbols

Name	Symbol	Use in Flowchart
Oval		Denotes the beginning or end of the program
Parallelogram		Denotes an input or output operation
Rectangle		Denotes a process to be carried out e.g. addition, subtraction, division etc.
Diamond		Denotes a decision (or branch) to be made. The program should continue along one of two routes. (e.g. IF/THEN/ELSE)
Flow line		Denotes the direction of logic flow in the program

Flow chart: example

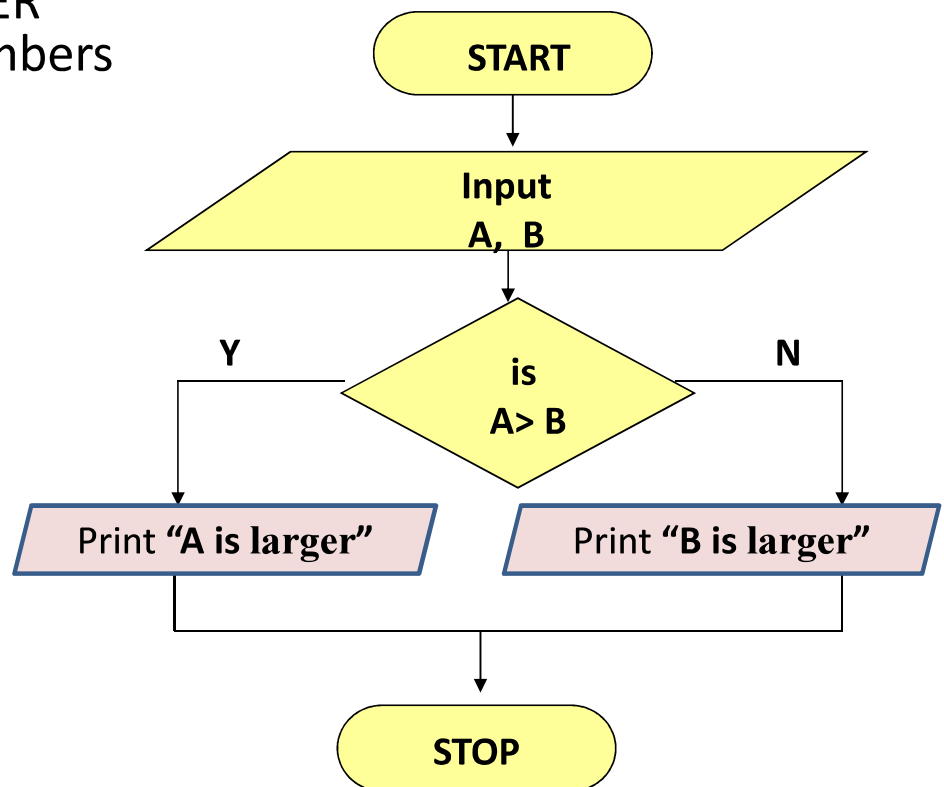


Algorithm: Average of three numbers

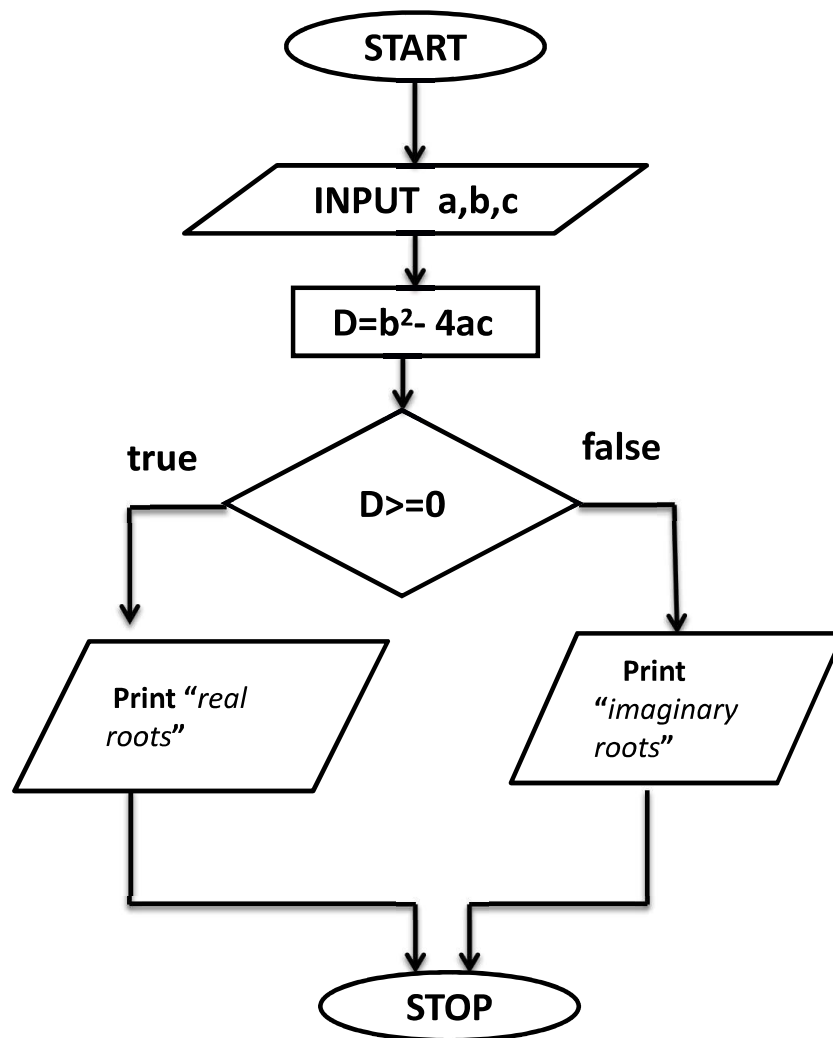
1. START
2. Input A,B and C.
3. $AVG = (A+B+C)/3$
4. Print AVG
5. END

Taking decision (if-else structure)

- Write an algorithm LARGER between two distinct numbers

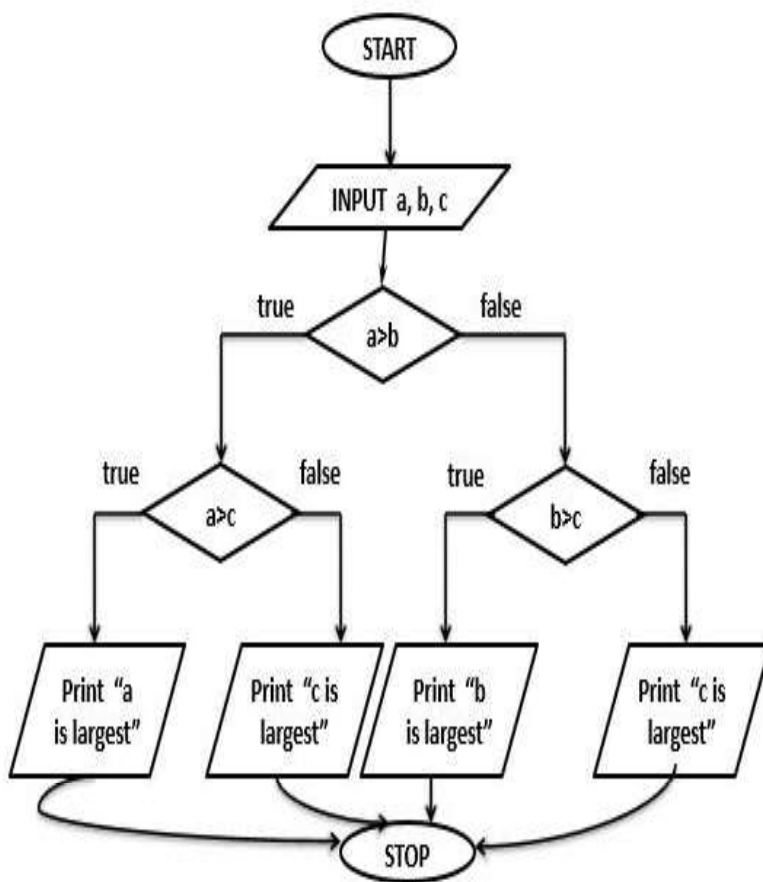


Flow chart to check that roots of a quadratic equation are real or imaginary.

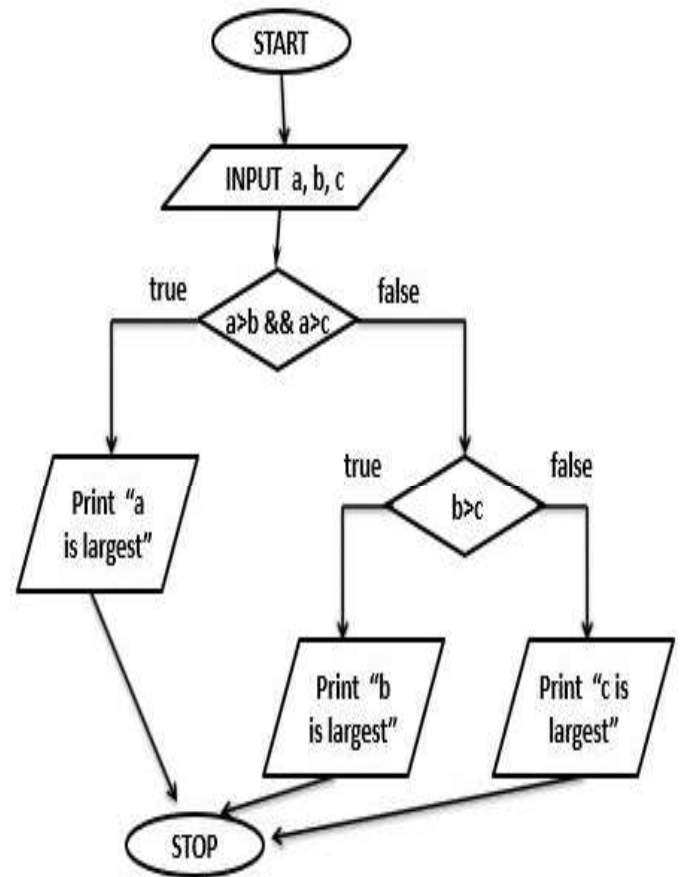


Decision Making in 'C'

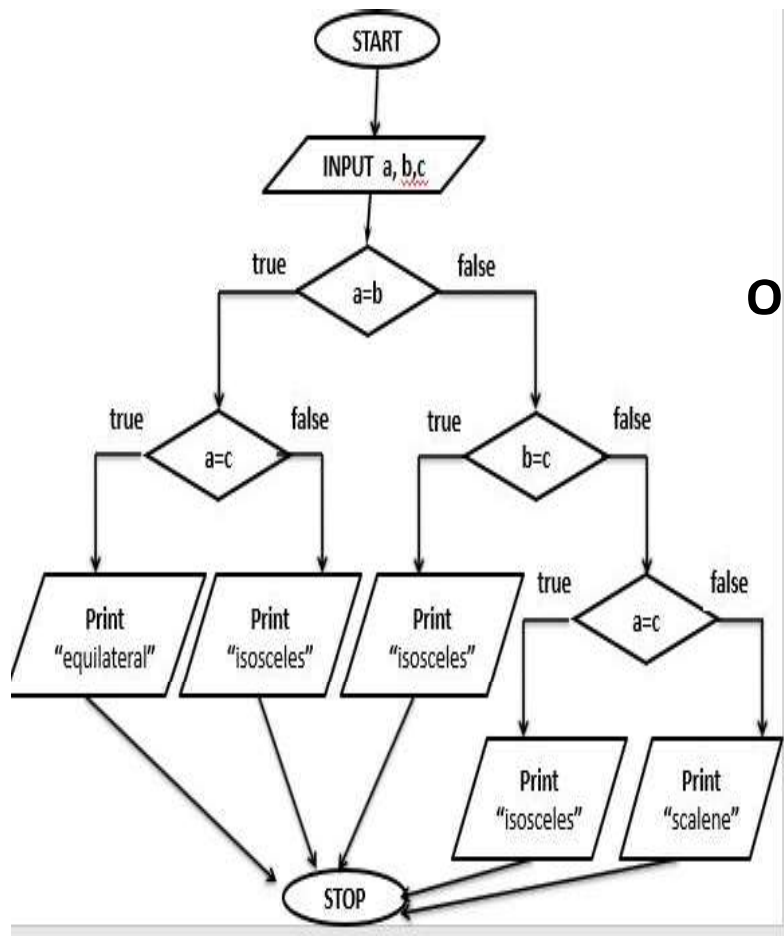
Example: Find the largest between three distinct numbers:



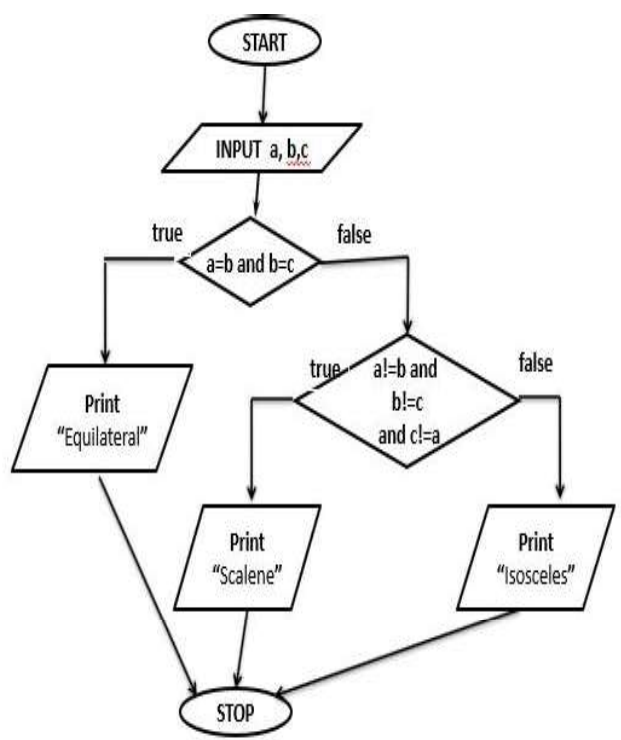
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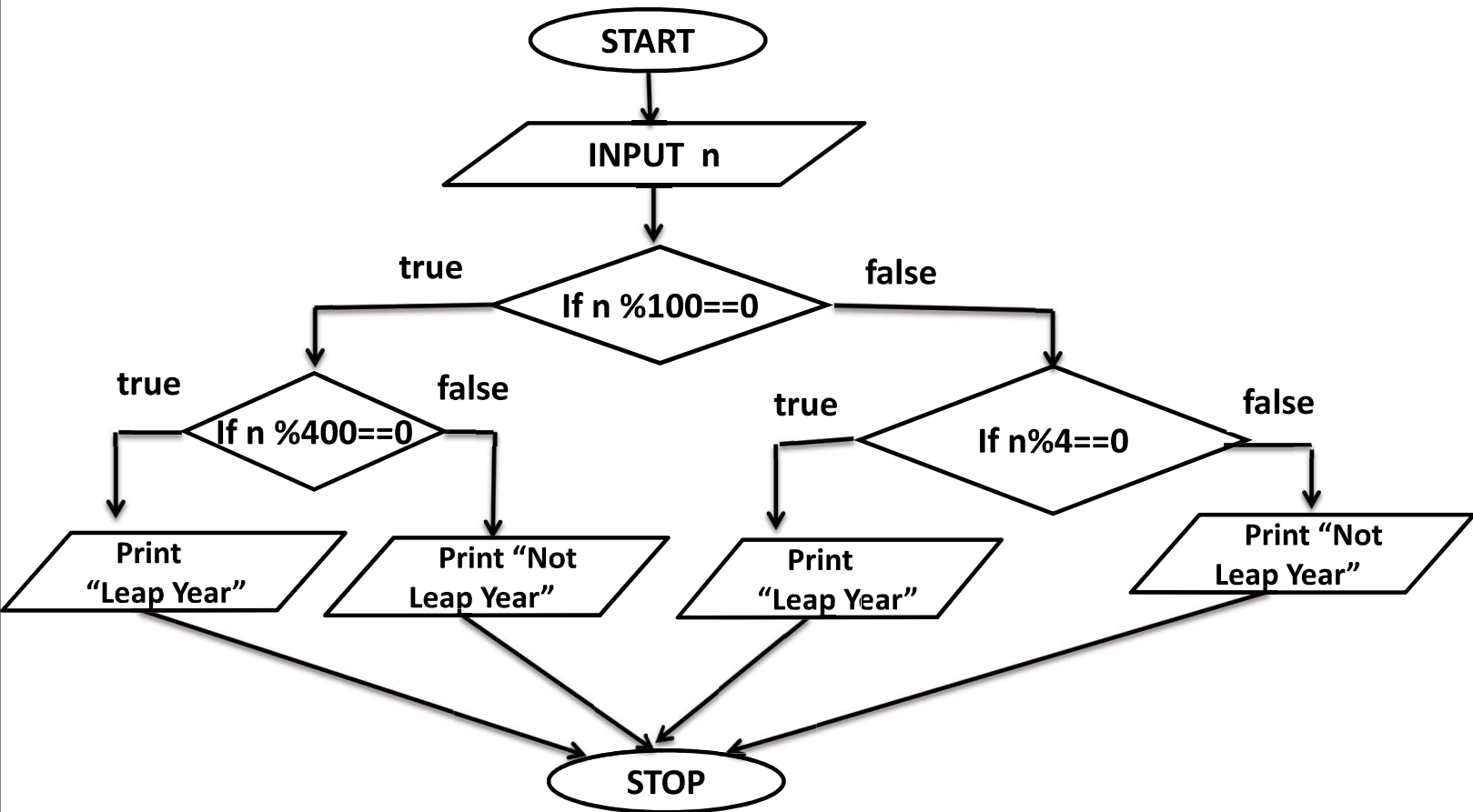
Example: check triangle is equilateral, isosceles or scalene



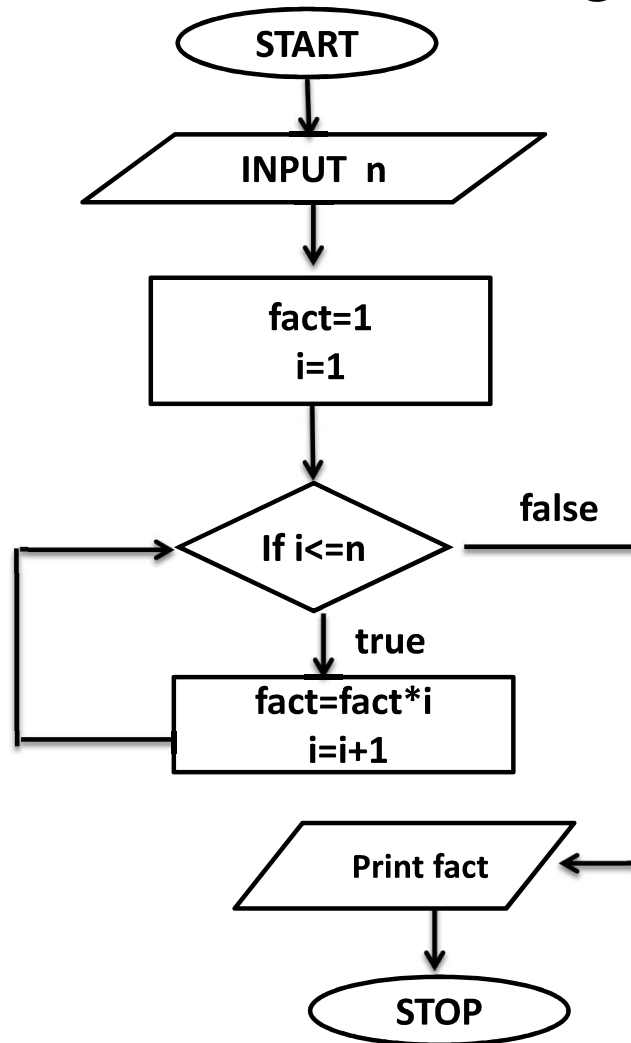
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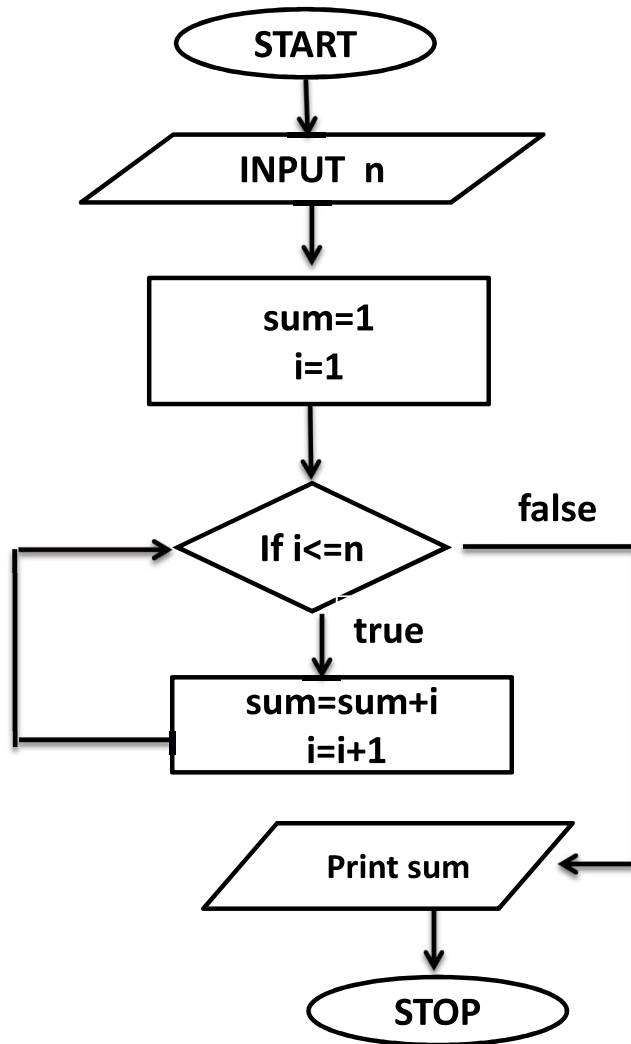
Flow chart to check leap year



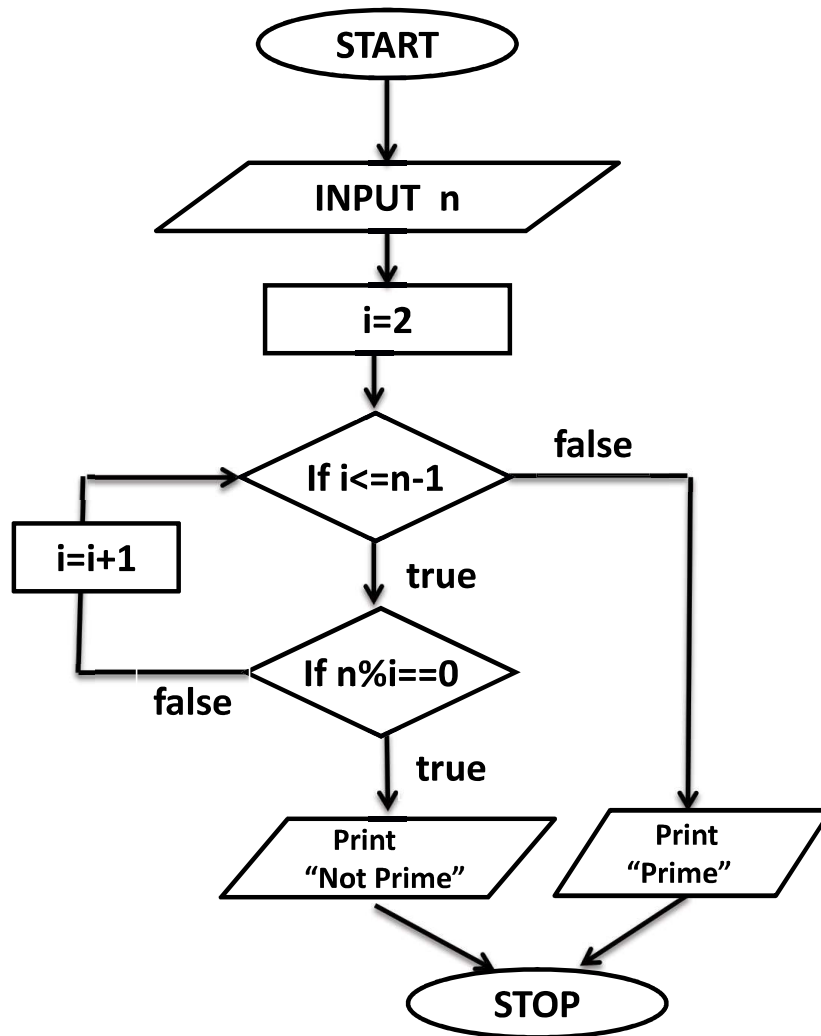
Flow chart to calculate factorial of a given number

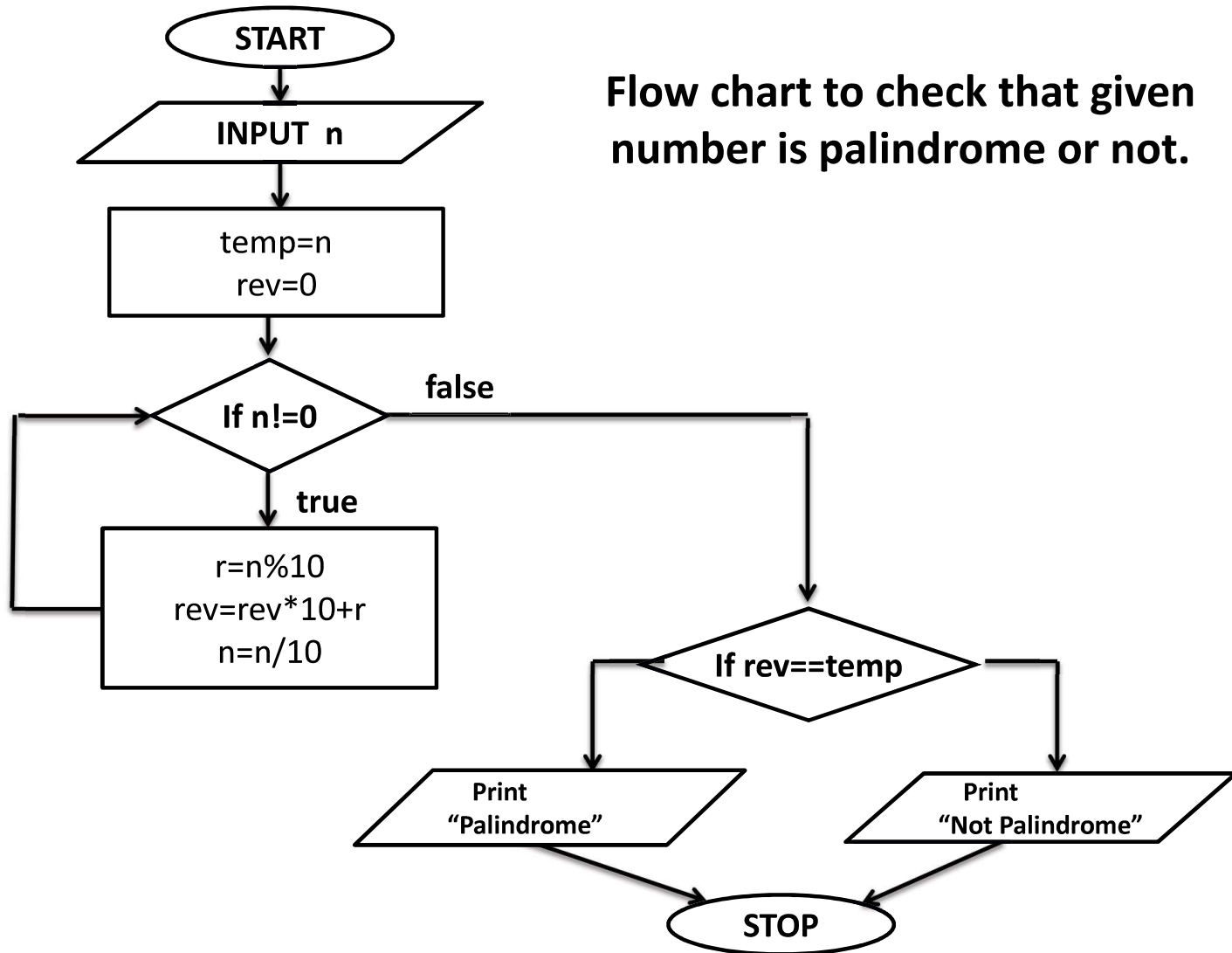


Flow chart to calculate sum of first n natural numbers



Flow chart to check that given number is prime or not





Flowchart to print Fibonacci series

